

⚠ THE PROBLEM

1. DRASTIC PRICE VARIANCE

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Major Mall Rate
\$20.00+

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Hidden Lot Nearby
\$3.95

2. RIGID SEARCH RESTRICTIONS

Parkopedia App: Require exact 6-digit Postal Code.
GoPark App: Accepts landmarks (Mall, Hospital, Building) + Full address.

👥 USER STORIES

- I want to search a landmark, avoiding postal code lookups.
- I want to input my duration to see the absolute cheapest options nearby.
- I want to sort by price vs walking distance to make informed decision
- I want to check live lot availability to avoid queueing at full malls

🛡 SE PRACTICES

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VERSION CONTROL
Git/GitHub tracking, branches for features, and PR code reviews before main merges

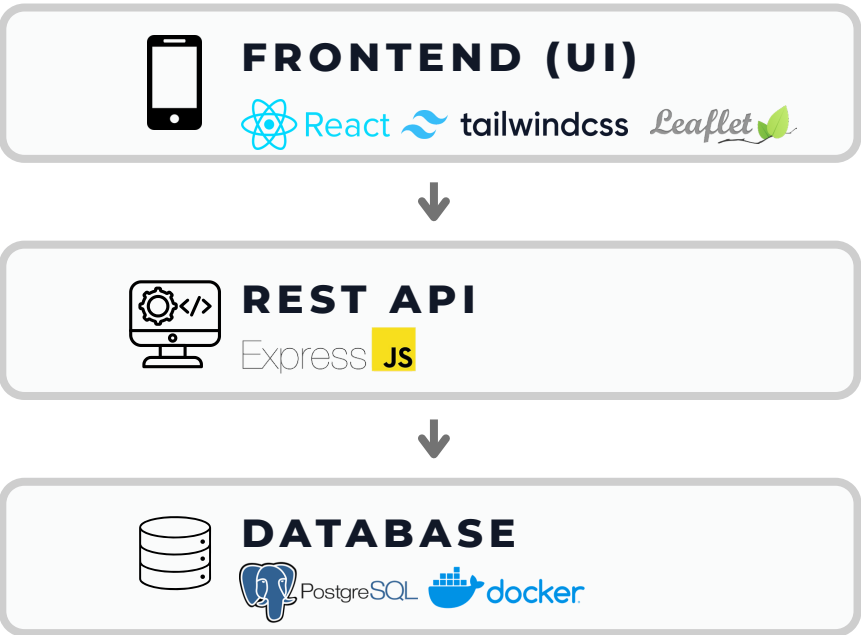
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AGILE DEV APPROACH
Features broken into small tasks, weekly syncs, and immediate feature testing.

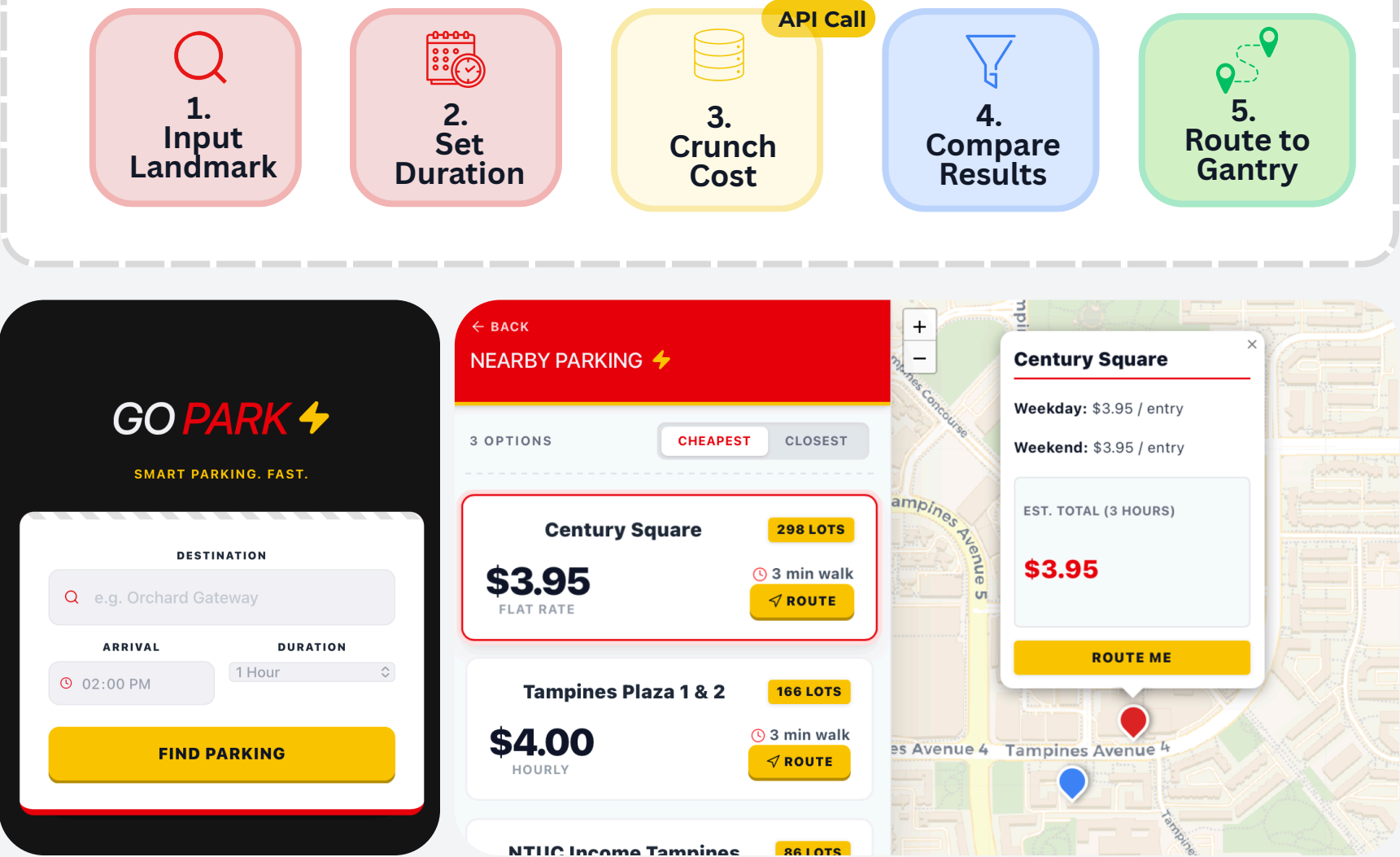
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SECURITY BASICS
Hidden API keys (.env), Bcrypt password hashing, and SQL injection sanitisation.

🏗 SYSTEM ARCHITECTURE



APP USER FLOW



CORE ENGINE MECHANICS

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1. LIVE AVAILABILITY TRACKING

Fetches live parking lot capacity data via HDB/LTA API Gateway to end the frustration of queueing at a full parking structure.

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2. DYNAMIC COSTING ALGORITHM

Calculates precise, final fees based on arrival time, estimated duration, peak surcharges, and hidden flat-rate rules that create massive savings.

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3. LANDMARK SEARCH

Translates natural language text (e.g. "Orchard Gateway") into direct GPS coordinates via integration with OneMap SG Geocoding API.

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4. ROUTE HANDOFF

Hand-off GPS coordinates directly to native Google Maps/Waze, routing the driver directly to the selected parking gantry entrance.

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5. INTERACTIVE FILTERING

Dynamic map visualization allows instant weight-sorting for "Cheapest Price" vs "Shortest Walking Distance" to make the optimal decision based on current needs.

🎬 FUTURE PLANS

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EV Integration: Allow users to filter parking options that guarantee available EV charging points.
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Grace Nudges: Push notifications reminding users their 10-minute grace period or flat-rate is ending soon.